

Exploring the Data

```
In [19]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
```

```
In [20]: df = pd.read_csv("data.csv")
df.head()
```

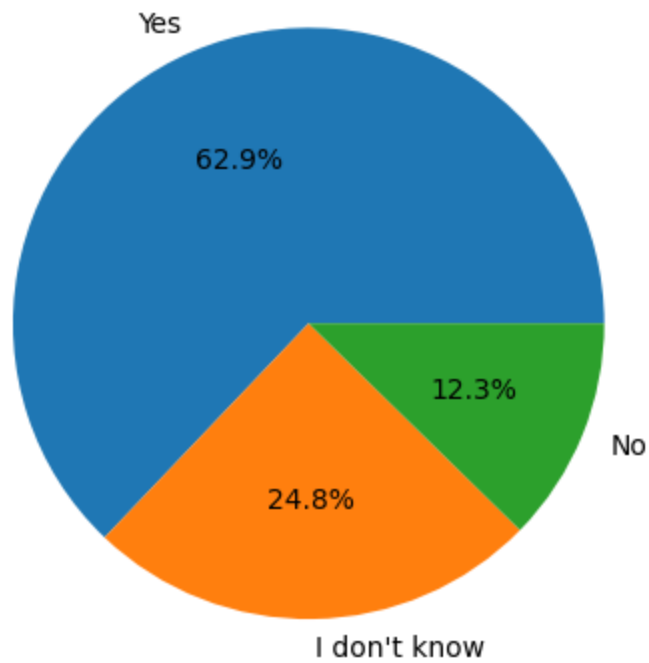
```
Out[20]:
```

	tech_company	benefits	workplace_resources	mh_employer_discussion	mh_coworker
0	Yes	No	I don't know		No
1	Yes	Yes	No		No
2	Yes	I don't know	No		Yes
3	Yes	Yes	I don't know		Yes
4	Yes	Yes	No		No

Preliminary visualizations

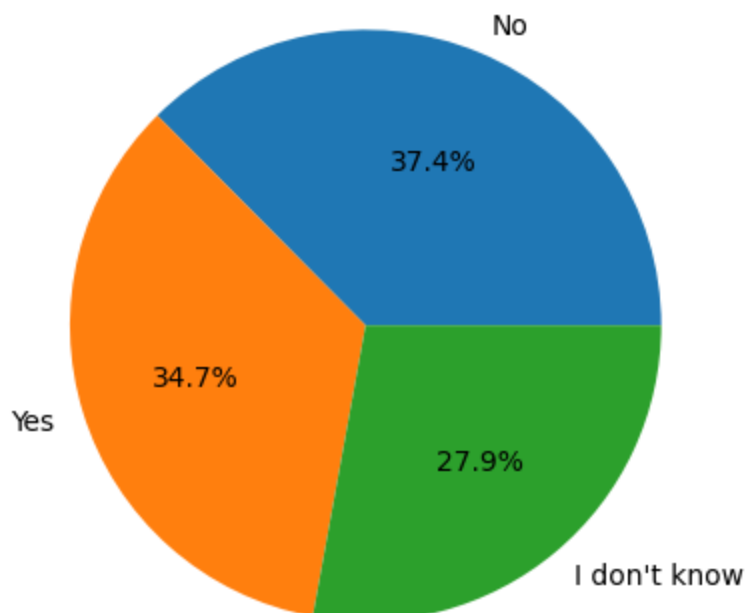
```
In [21]: benefits = df["benefits"].value_counts()

plt.pie(benefits, labels=benefits.index, autopct="%1.1f%%")
plt.show()
```



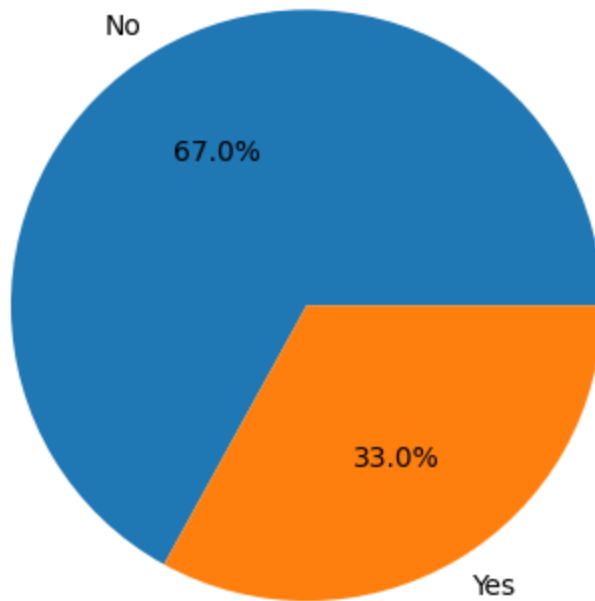
```
In [22]: resources = df["workplace_resources"].value_counts()

plt.pie(resources, labels=resources.index, autopct="%1.1f%%")
plt.show()
```

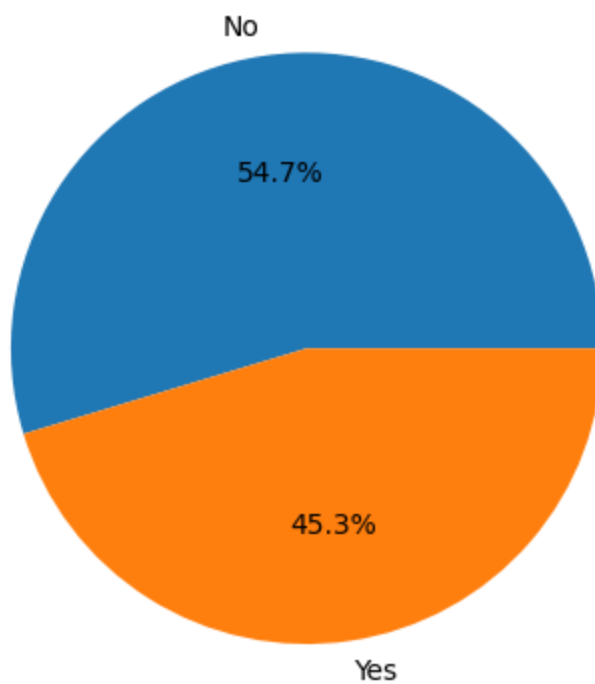


```
In [23]: employer_discussion = df["mh_employer_discussion"].value_counts()
```

```
plt.pie(employer_discussion, labels=employer_discussion.index, autopct="%1.1",  
plt.show()
```

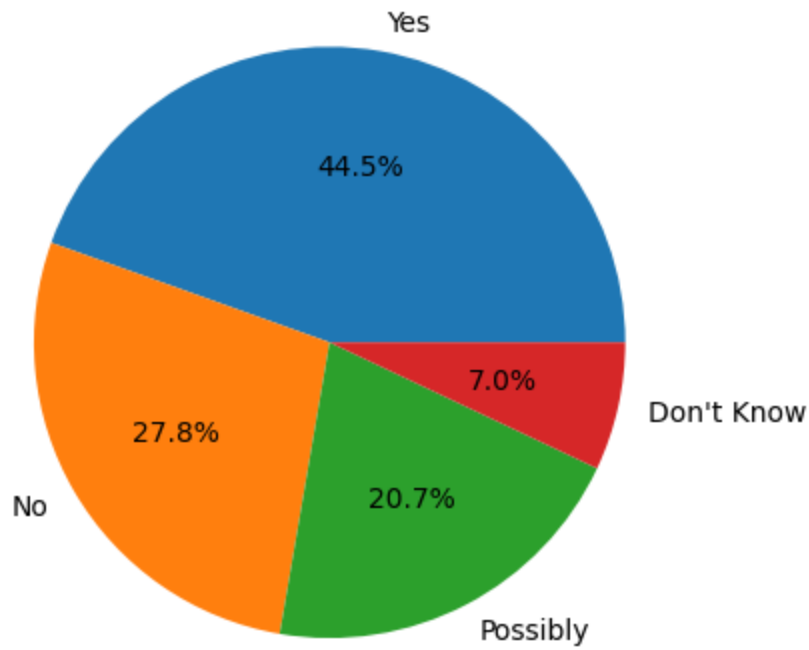


```
In [24]: coworker_discussion = df["mh_coworker_discussion"].value_counts()  
  
plt.pie(coworker_discussion, labels=coworker_discussion.index, autopct="%1.1",  
plt.show()
```

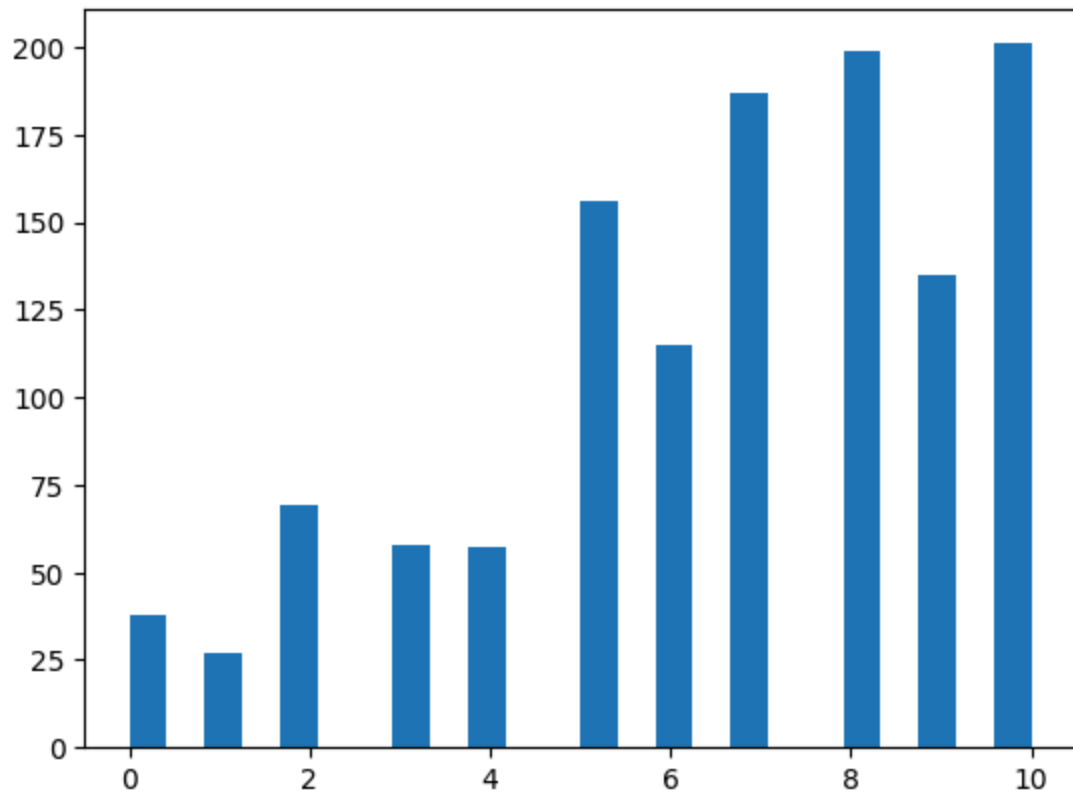


```
In [25]: mental_health = df["mental_health"].value_counts()

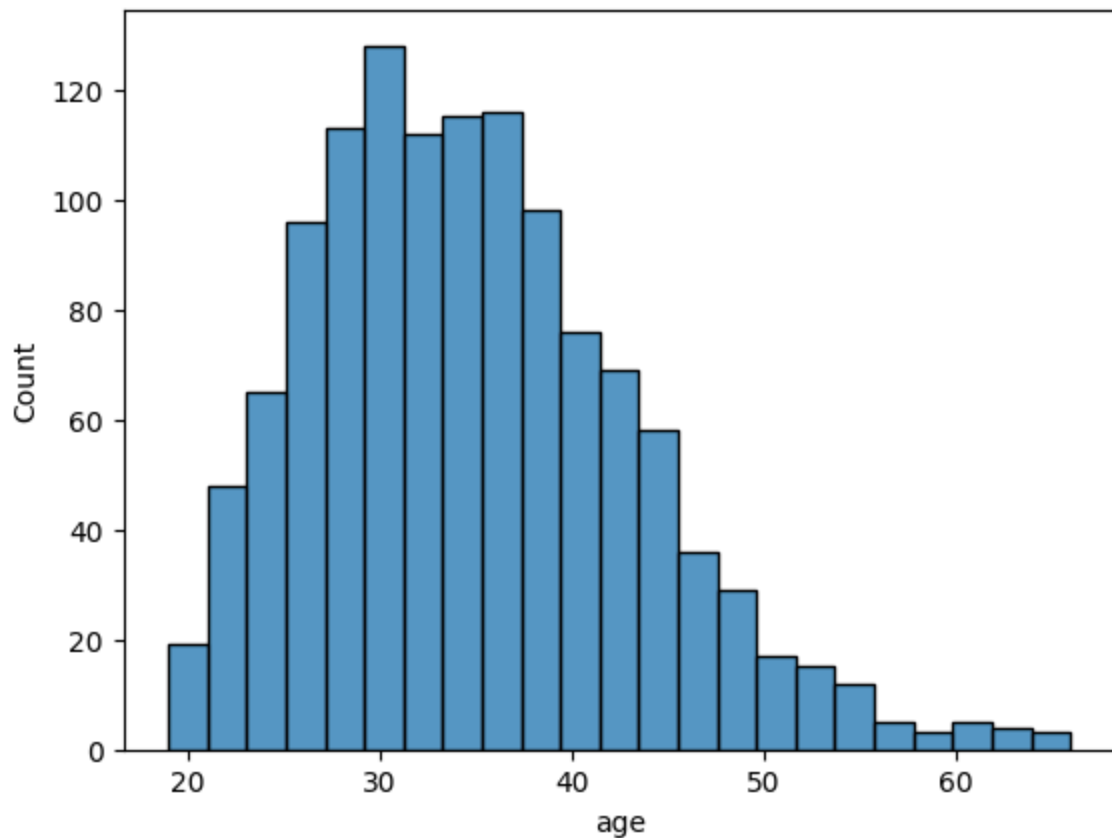
plt.pie(mental_health, labels=mental_health.index, autopct="%1.1f%%")
plt.show()
```



```
In [26]: plt.hist(df['mh_share'], bins=24)
plt.show()
```

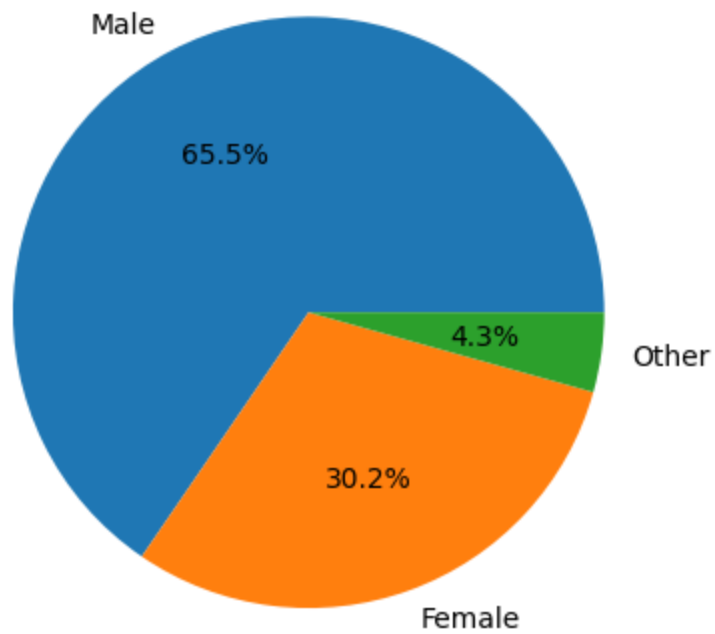


```
In [27]: sns.histplot(df, x="age")  
plt.show()
```

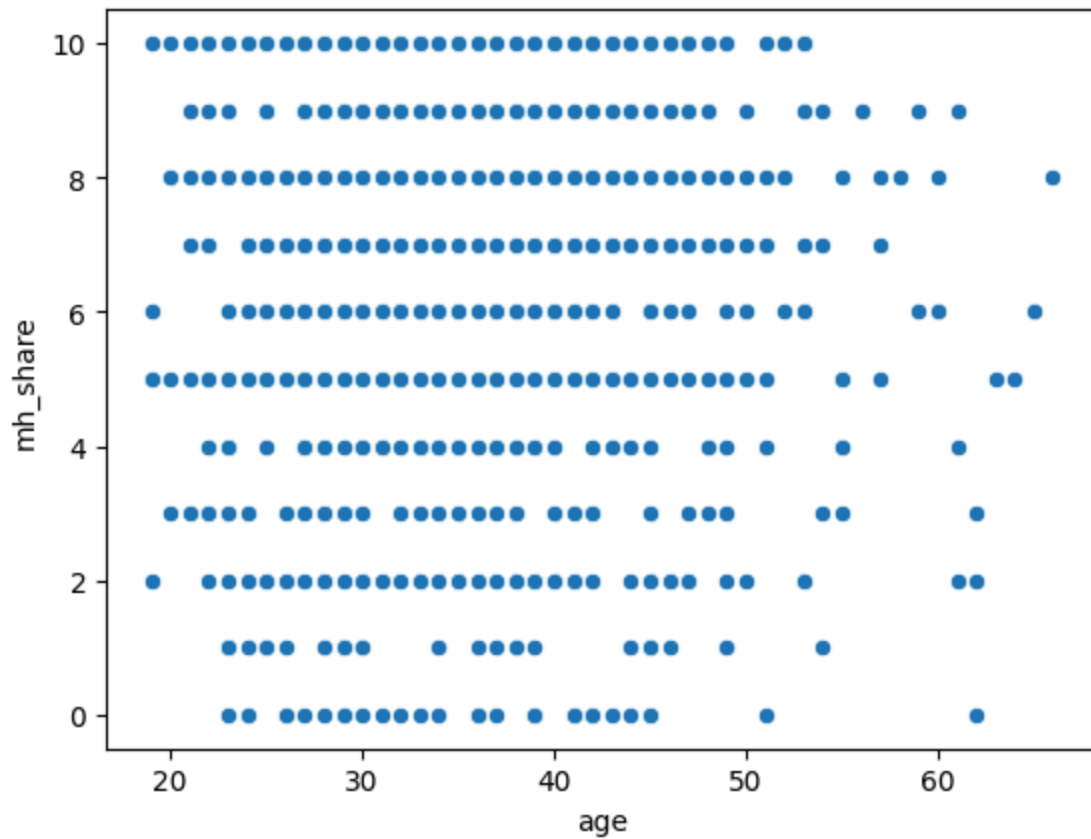


```
In [28]: gender = df["gender"].value_counts()

plt.pie(gender, labels=gender.index, autopct="%1.1f%%")
plt.show()
```



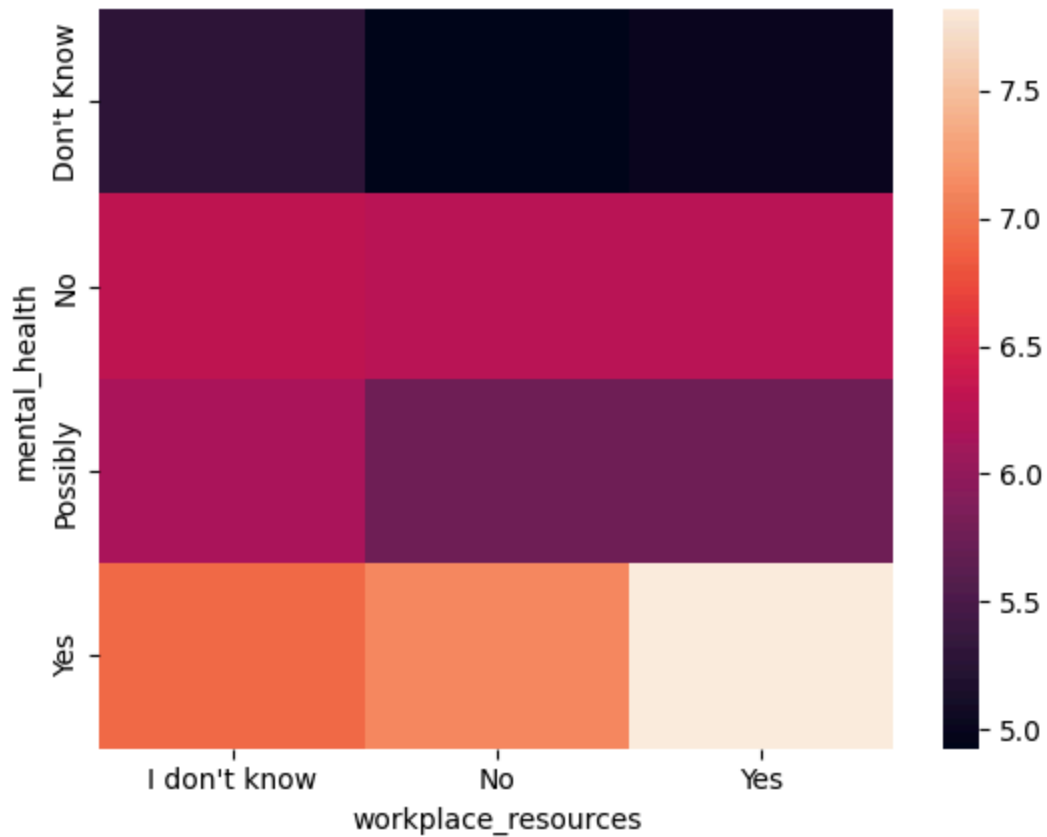
```
In [29]: sns.scatterplot(df, x="age", y="mh_share")
plt.show()
```



Data transformations

```
In [30]: table = df.pivot_table(index="mental_health", columns="workplace_resources",  
sns.heatmap(table)
```

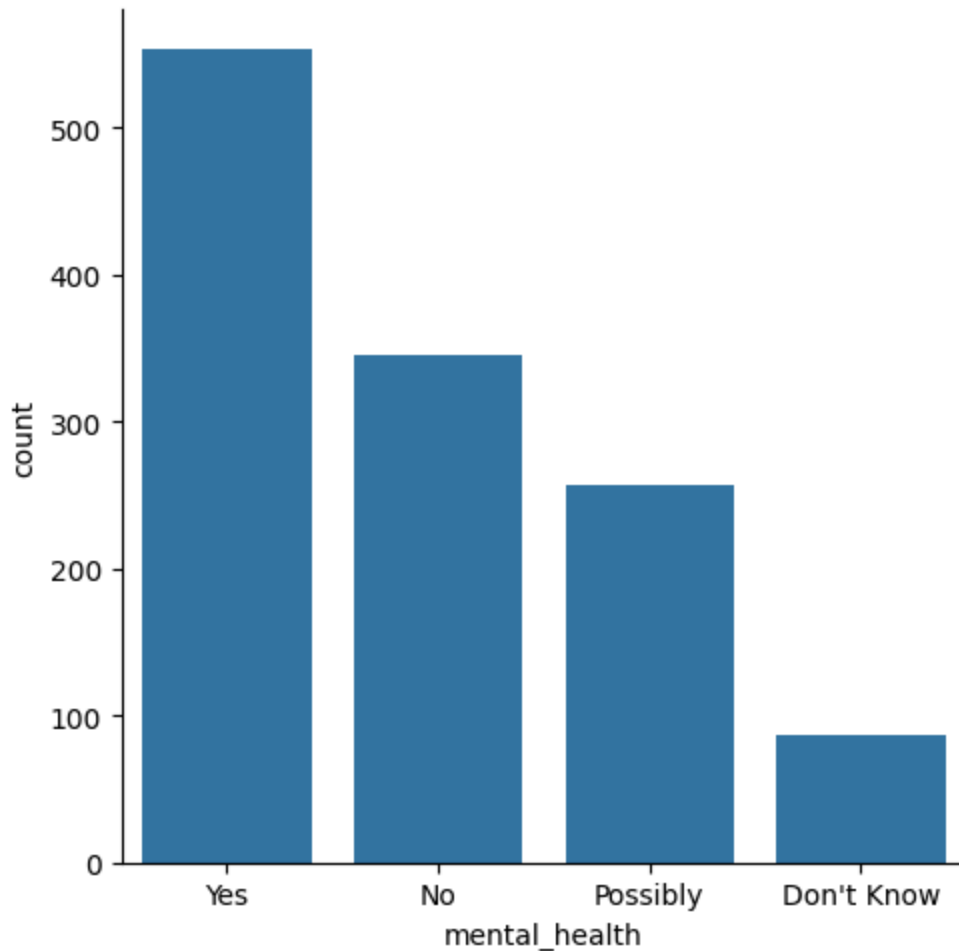
```
Out[30]: <Axes: xlabel='workplace_resources', ylabel='mental_health'>
```



```
In [31]: mental_health = df["mental_health"].value_counts().reset_index()
mental_health.columns = ["mental_health", "count"]

sns.catplot(mental_health, x="mental_health", y="count", kind="bar")
plt.show
```

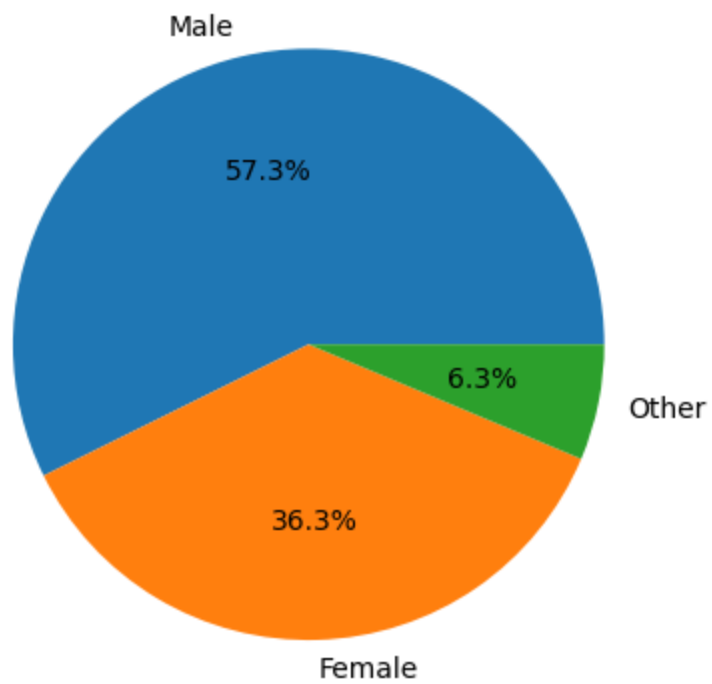
```
Out[31]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [32]: male_with_mental_health = df[(df["mental_health"] == 'Yes') & (df["gender"] == 'Male')]
female_with_mental_health = df[(df["mental_health"] == 'Yes') & (df["gender"] == 'Female')]
other_with_mental_health = df[(df["mental_health"] == 'Yes') & (df["gender"] == 'Other')]

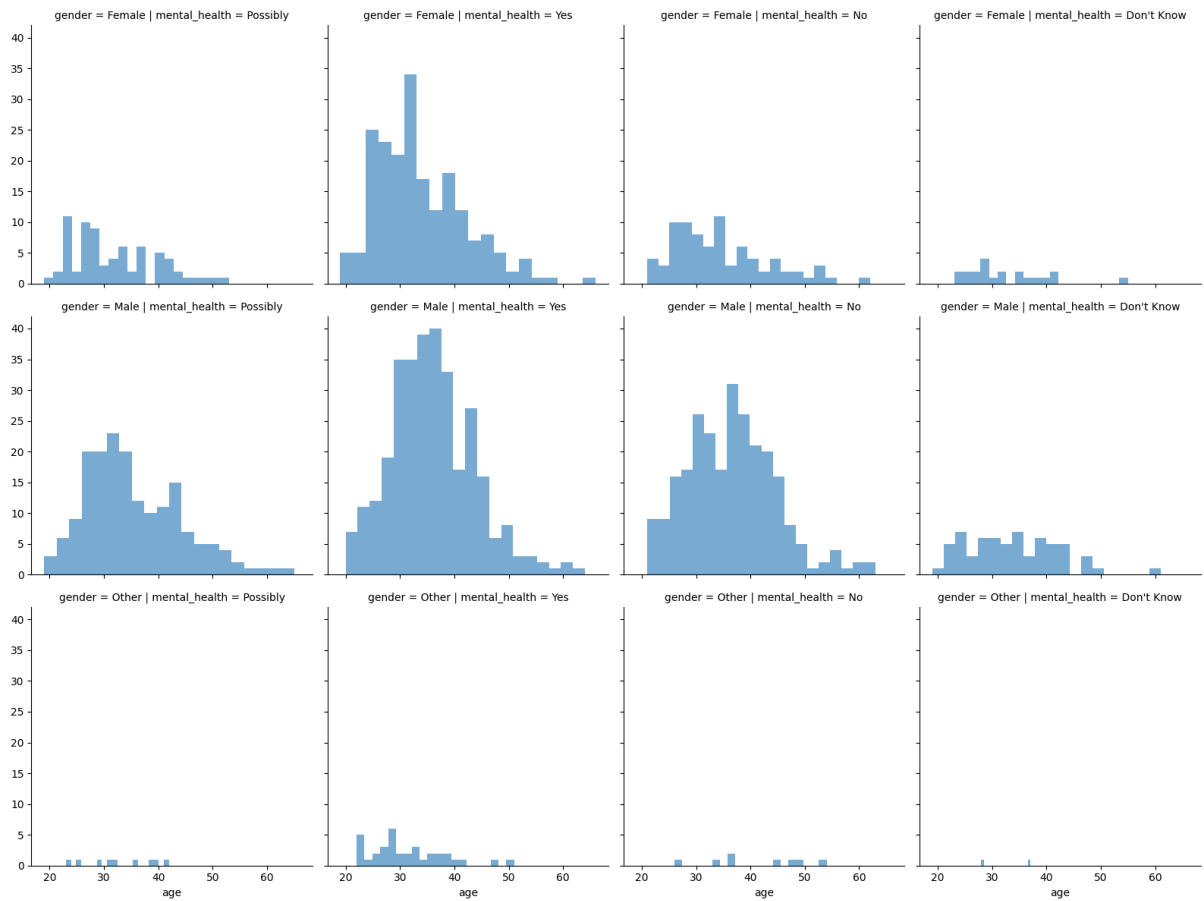
sizes = [
    male_with_mental_health['gender'].shape[0],
    female_with_mental_health['gender'].shape[0],
    other_with_mental_health['gender'].shape[0]
]
labels = ['Male', 'Female', 'Other']

plt.pie(sizes, labels = labels, autopct = '%1.1f%%')
plt.show()
```



Different plot types

```
In [33]: g = sns.FacetGrid(df, row = 'gender', col = 'mental_health', height = 4)
g.map(plt.hist, 'age', bins = 20, alpha = 0.6)
g.add_legend()
plt.show()
```



```
In [34]: px.parallel_categories(df,
    dimensions=["gender", "workplace_resources", "mental_health"])
```

